

Right on target: monopods and polystyrene spheres as an alternative to traditional tripods for terrestrial laser scanning

Léo Jourdan, Kennedy Korkola , Paul Piascik, and Patrick M. A. James

Institute of Forestry and Conservation, John H. Daniels Faculty of Architecture, Landscape and Design, University of Toronto, Toronto, ON M5S 3B3, Canada

Corresponding author: **Kennedy Korkola** (email: kennedy.korkola@mail.utoronto.ca)

Abstract

Terrestrial lidar is an increasingly popular tool to understand forest stand structure, biomass distribution, and fuel characteristics. A common approach is a multiscan protocol, which reduces the risk of occlusion. To merge the different scans, identical reference points (targets) are necessary. However, the additional equipment required can be expensive and cumbersome. We tested three different target systems during field work in Ontario's Boreal forest and compared each system's cost, weight, precision, and ease of use. We found that currently used target configurations are typically expensive and heavy, but straightforward to use. We proposed an alternative target configuration, a monopod, which was inexpensive, lightweight, and more precise; however, it was not suitable in all terrains. Our goal was to identify cost-effective and weight-efficient equipment alternatives to make lidar data acquisition more accessible.

Key words: lidar, remote sensing, fieldwork, equipment, TLS

Introduction

Forest monitoring and inventory are essential components of forest management, but they often rely on costly and time-consuming manual sampling (Duff et al. 2017). The collection of forest inventories is further complicated by the many logistical challenges associated with remote sampling locations (Pitt et al. 1997). In recent decades, numerous remote sensing approaches have been developed to improve forest monitoring by increasing the spatial extent of the surveyable area and reducing costs (Xiao-rui et al. 2005; Saatchi et al. 2007; Ferraz et al. 2009). These advances in remote sensing and satellite imagery allow for the collection of high-resolution data in unprecedented quantity and quality. For example, by calculating the time of return of high-frequency laser pulses, lidar technology enables 3D mapping of an area from which many fundamental forest attributes (e.g., canopy height, tree diameter, tree species) can be derived (Kelly and Di Tommaso 2015). However, these new technologies necessitate additional considerations for their data collection to be efficient and accurate.

One common and well-established variant of lidar is Terrestrial Lidar Scanning (TLS). TLS scanners are typically installed on a tripod and remain stationary while in operation. TLS is a proven data collection method in the field of forestry (Hopkinson et al. 2004), as its mapping of three-dimensional space is particularly useful for stand inventory assessment (Liang et al. 2016), biomass calculations (Calders

et al. 2015), and fuel estimation (García et al. 2011). While using TLS for data collection and processing requires time, computational power, and expertise, TLS is an excellent alternative to manual vegetation sampling as it can reduce sampling time and is less prone to human error (Liang et al. 2016). Despite their flexibility and convenience, TLS systems remain expensive and can be challenging to deploy in the field due to their weight and the additional equipment they require. Many opportunities exist to improve TLS field protocols to reduce their financial and logistical costs, as well as to increase the quality and utility of the resulting data.

The Leica BLK360 TLS scanner has emerged as a popular lidar tool because of its relative affordability, ease of use, and portability (Pokswinski et al. 2021; Tao et al. 2021; Kükenbrink et al. 2022). A well-known challenge associated with TLS-based approaches in stand mapping and inventory is occlusion, in which trees, rocks, and other vegetation can block laser pulses from a single scan. To overcome challenges posed by occlusion, TLS protocols typically adopt a multiscan approach by distributing scanning locations throughout the plot (e.g., Hillman et al. 2021; Boucher et al. 2022).

While the use of multiple complementary scans can help overcome problems related to occlusion, the resulting scans need to be aligned and merged to form a single point cloud of the area of interest. Aligning multiple scans in a complex forest environment is challenging as there are limited reference

points in the organic natural environment. This alignment challenge is distinct from alignment challenges in the built environment, where relatively clear, linear, and human-made features can be used to facilitate scan alignment. In the absence of clear features for scan alignment, researcher-placed targets that are visible from different scans can serve as the needed spatial reference points. Targets and their placement are an essential part of a multiscan TLS protocol as they ensure that the point cloud assembled from the individual scans is an accurate three-dimensional map of the area. Most lidar scanners, including the Leica BLK 360, do not use integrated GPS (Tremblay and Béland 2018). As such, visual markers are one of the only effective ways to align multiple scans.

Targets for lidar point cloud alignment come in various shapes, sizes, and configurations. Manufacturers sell two main types of targets: “Black/White” (B&W) and spherical. The B&W targets are flat and have two opposing black squares, forming a crosshair at the center of the target. These targets are typically installed on a tripod to ensure that they do not move during the multiscan process and that they can be rotated as needed—since B&W target’s surface must be facing the scanner. Spherical targets are made of highly reflective materials and do not need to be rotated during the multiscan procedure. Placing and rotating targets between scans is a time-consuming task (Tremblay and Béland 2018). In addition to manufactured targets, home-made targets can replicate the B&W and spherical ones for lower costs. For example, the crosshair B&W pattern can be printed on a flat and rigid surface that can be affixed to the top of a tripod. Spherical targets can be replicated with plastic or polystyrene balls (Tao et al. 2021). These targets can be installed on different stands that ensure they do not move during the scanning process. Different types of tripods can be used for this, as well as monopods, which are poles that are staked into the ground. A target’s location must remain constant throughout the multiscan process to ensure a proper alignment of the multiple point clouds. Therefore, it is imperative to ensure the stand chosen is stable and properly secured so that it will not be impacted by wind.

Forestry research is often conducted in remote locations, which requires long days in the field, physical exertion from carrying equipment, and bushwhacking through dense vegetation. Therefore, equipment must be as light and compact as possible to reduce strain on personnel and keep crew morale high. Also, due to the remote nature of the work, repair and replacement options for gear must be accessible. Forestry field research is also often carried out with modest budgets, and affordable solutions for high-quality data collection are desirable. In this research note, we evaluate several different scanning target configurations in terms of ease and efficiency of application while also ensuring that we captured data of sufficient quality. Our goal was to establish which combination of TLS scanning target configuration was most appropriate for a forestry context in terms of stability, weight, cost, and ease of use in the field. Our recommendations aim to reduce the barriers to the uptake of TLS scanning in forestry studies.

Materials and methods

Study region

We tested different target configurations in various forest types within the Ontario shield. We sampled 32 sites in the northeast around North Bay, Timmins, and Kapuskasing, and 48 in the northwest around Dryden and Red Lake (Fig. 1). Sites were located a maximum of 1 km from a road or trail and typically required 5–30 min of hiking to access. These sites were characterized by mostly mixedwood and conifer forest types. Dominant tree species for the eastern sites included black spruce (*Picea mariana*), white spruce (*Picea glauca*), balsam fir (*Abies balsamea*), jack pine (*Pinus banksiana*), white birch (*Betula papyrifera*), trembling aspen (*Populus tremuloides*), and balsam poplar (*Populus balsamifera*). Important understory species included green alder (*Alnus alnobetula*), beaked hazel (*Corylus cornuta*), and wild raspberry (*Rubus idaeus*). In the northwest, sites were established around Dryden and Red Lake. The dominant overstory species included jack pine, black spruce, balsam fir, white birch, trembling aspen, and balsam poplar. The understory was relatively clear in these sites, with the forest floor being dominated by red-stemmed feather moss (*Pleurozium schreberi*) and lowbush blueberry (*Vaccinium augustifolium*). Sites varied in their amount of downed woody debris and understory growth due to previous disturbance, which presented increased challenges to movement to and within the stand, thus motivating research into lightweight and affordable solutions to establish an effective scanning protocol.

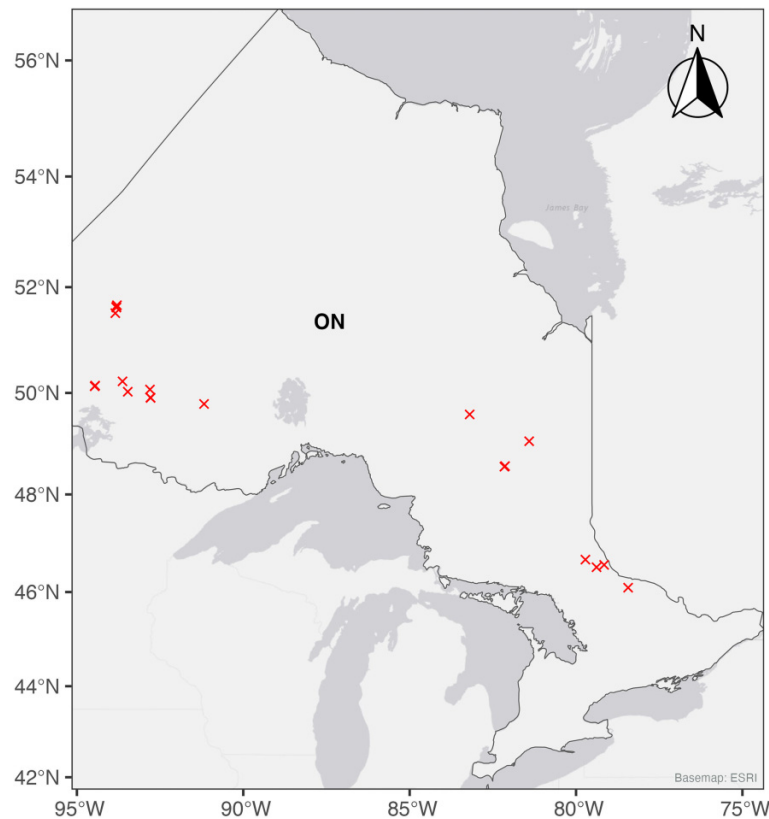
Materials

We used the Leica BLK360 to test the effectiveness and efficiency of different target configurations using a multiscan protocol in forest stands (Boucher et al. 2022). This protocol requires eight stationary targets and nine scans from different locations within the plot. The different target configurations were compared using their overall weight per unit, the cost of assembly, and ease of use in the field and across varying terrains.

Our three configurations varied in terms of the target type (i.e., flat or spherical) and target mount (i.e., tripod or monopod). The first configuration was a Leica tripod (Leica, SKU #670223) with Leica B&W targets (Leica, SKU #822694) (L), the second was a land surveying tripod with a black and white target printed on corrugated plastic sheets (LS), and the third was a telescopic monopod (Woods, SKU #0766642) with a polystyrene sphere target (MP) (Figs. 2 and 3).

The L configuration required no assembly, as every component was purchased directly from Leica (<https://leica-geosystems.com/>). We made a derivative of the L configuration for the LS configuration: we glued a piece of wood to the top of a land survey tripod, on top of which we attached a screw clamp. The clamp holds an adjustable telescopic steel awning pole (Fig. 4). The corrugated plastic sheet target is fixed to the end of a polyvinyl chloride pipe, which sits atop the steel pole. Finally, our monopod (MP) configuration included a telescopic aluminum awning pole with a stake made from a whittled branch. The awning pole had a small metal rod at the top,

Fig. 1. Site locations in which the different target configurations were tested. The sites were located across Ontario's Boreal forest. Sites varied in understory density and downed woody debris. Figure was created using R v4.5.1 (R Core Team 2025) using the World Light Gray basemap from ESRI.



which we used to puncture a hole in the polystyrene sphere target.

Scanning protocol

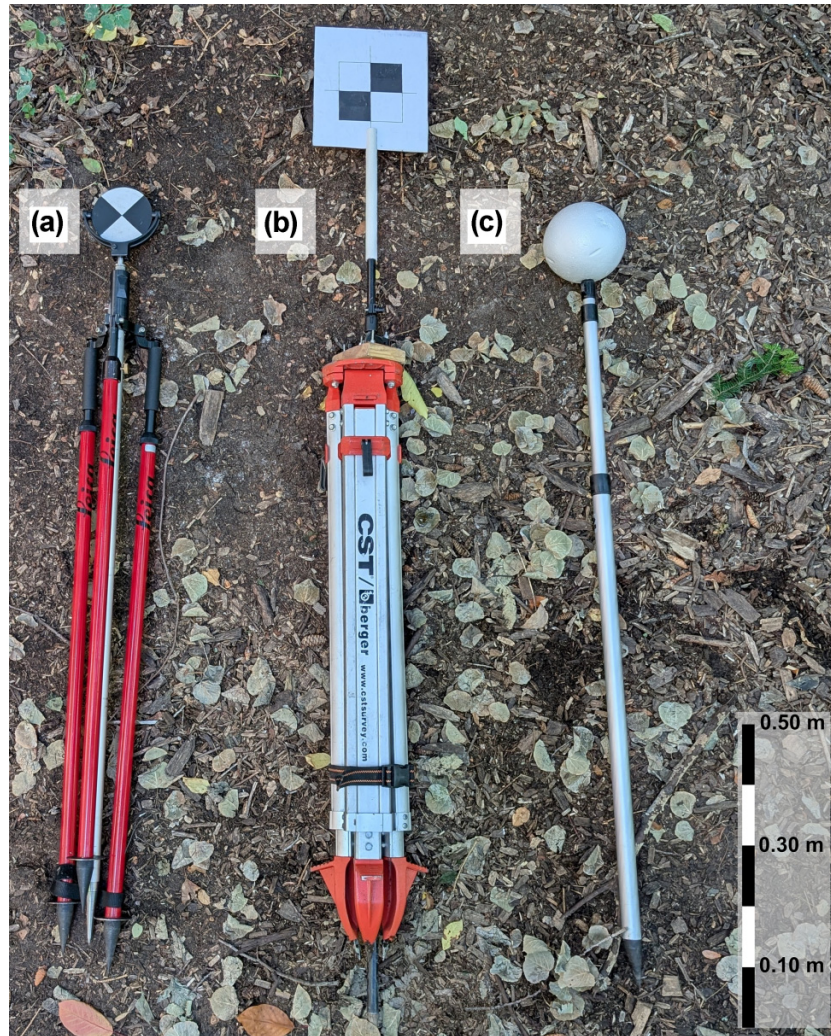
The location of each site center was established randomly within each stand of interest. We completed nine TLS scans total for each site as outlined in Fig. 5 (Boucher et al. 2022). The scans were initiated using the Leica Cyclone Field 360 software on an iPad, connected to the scanner via Wi-Fi. Around the plot center, eight targets were set up, one in each cardinal and intercardinal direction. Cardinal targets were set up 12 m from the plot center and intercardinal were positioned 7 m from center. When placing the target at the exact distance was not possible, it could be moved within 1 m radius. Scanning began at the center of the plot, followed by the north direction and continuing clockwise around the plot to end at northwest. For the flat black and white targets, the targets were rotated after each scan to face the new scanning location. Spherical targets did not need to be rotated. Immediately following a scan, at least one of the targets was registered on the 360 imagery using the Leica Cyclone Field 360 software to ease the alignment of the scans and maintain directionality during data processing.

Data processing

Data were transferred from the iPad to our field laptop via ethernet connection through the Leica Cyclone REGISTER

360 PLUS (BLK Edition) software. Once the data were transferred, we registered all additional targets in each scan and then aligned the nine scans for each site. The targets were either registered as black and white targets or as spherical targets, depending on what was used in the field. When using spherical targets, we auto-aligned the scans using the target alignment. With the B&W targets, we manually aligned the targets and then shifted the scans using the visual comparison tool. We compared target positional precision between the B&W targets used in the L and LS configurations and the spherical targets used in the MP configuration by calculating the root mean square (RMS) of observed target position from the northwest sites in Dryden and Red Lake. These sites have sparser understory, which allowed us to better isolate the target precision of the different configurations. For each site, we extracted the observed target positions according to each of the nine scans after alignment. We removed any target observation that was further than 1 m from that target's medoid position (i.e., the position of the observation with the least distance from other observations), to remove any mislabelled targets. For a given target, the RMS calculation allowed us to measure how much the scans agreed on the position of a given target. Higher values of RMS suggest lower target positional precision and correspond to higher point cloud errors. A one-way analysis of variance (ANOVA), followed by a post hoc Tukey's test with $\alpha = 0.05$, was used to test for significant differences in RMS between the L, LS, and MP designs.

Fig. 2. The three target configurations in their semicollapsed position, where (a) is the Leica, (b) is the Land Survey, and (c) is the Monopod configuration. Photo taken by Kennedy Korkola.



Results

We used the BLK360 TLS to test different target configurations across northern Ontario. We compared these target configurations in terms of their cost, weight, precision, and stability in different terrains (Table 1).

Leica tripods (l)

Leica tripods paired with Leica black and white (B&W) targets (Fig. 3a) were the most expensive option. Each tripod cost \$550.00 CAD and each target was \$440.00 CAD from Leica Geosystems (leica-geosystems.com). The weight of the L configuration totalled 3.83 kg, which contributed to their stability in the field but also made it difficult to carry through dense or uneven terrain. However, the L system was relatively compact when packed, as one tripod is 0.13 m across at its widest but tapers down to 0.06 m (Table 1).

While Leica tripods are costly and heavy, they can be set up and stabilized very swiftly, even when extending to their maximum height, because of their built-in level bubble and thumb release leg locks. The high-quality construction of the

tripods and targets offered increased accuracy during scan alignment but required turning of targets between each scan. On average, a site would take 2 h to complete, with nine scans at 13 min each. The TLS would complete the scan in 6 min 10 s, and the remaining 6 min 50 s would be used to move the scanner and turn the targets.

Land survey tripods

The LS tripod configuration, consisting of a telescopic pole and corrugated plastic B&W target (Fig. 3b), was less expensive than using the L tripods. The LS tripod could be purchased for \$97.00 CAD from ITM Instruments (www.itm.com). In addition to their affordability, their widespread use in forestry makes this option likely more accessible for researchers. The steel telescoping pole was purchased for \$15.99 CAD from Canadian Tire (www.canadiantire.ca). Additional securing hardware was purchased from local hardware stores, and the targets could be made for as low as \$2.00 CAD. However, these tripods were heavy, due to the combined weight of the tripod and telescoping pole, weighing 4.55 kg per unit. These tripods were also wide when packed,

Fig. 3. The three target configurations set up and deployed where (a) shows the Leica configuration, (b) shows the Land Survey configuration, and (c) shows the Monopod configuration. Photo taken by Léo Jourdan.

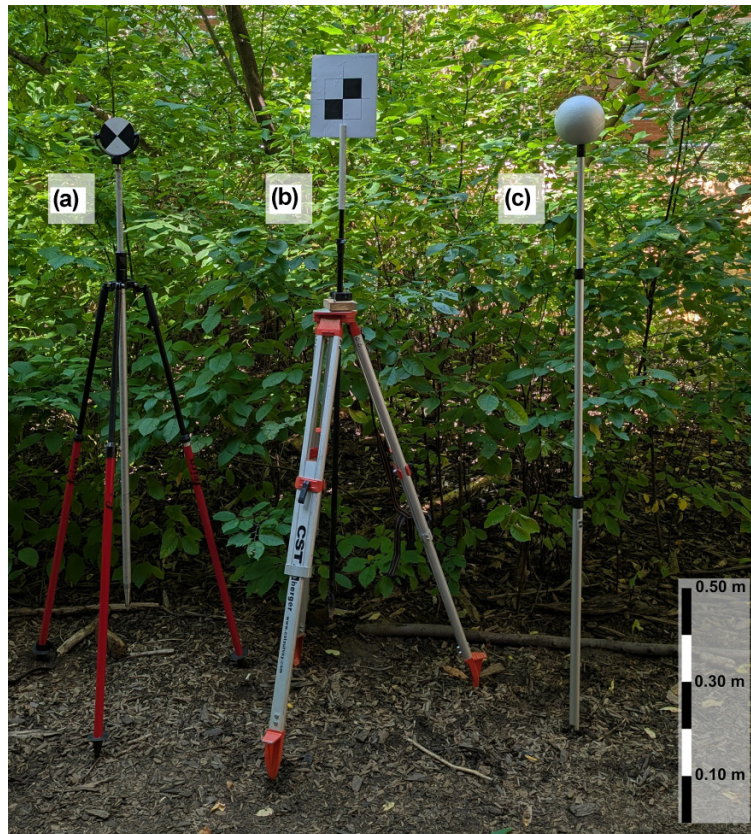
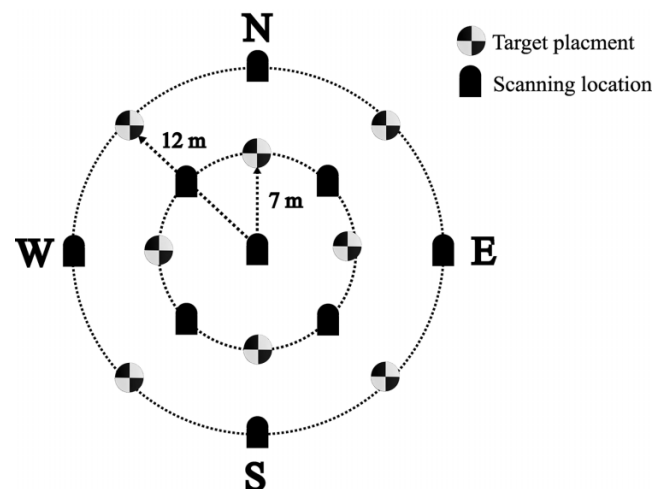


Fig. 4. Close up of the Land Survey tripod configuration. The wood block and screw clamp were used to secure the steel telescoping awning pole in the centre. Photo taken by Léo Jourdan.



Fig. 5. Plot layout for the fuel sampling protocol (Boucher et al. 2022). Each plot is scanned nine times: in the centre, at the cardinal, and the intercardinal directions.



with a width of 0.13 m. They were relatively easy to set up in the field, particularly in open plots and in rocky terrain. However, their bulk and weight made them challenging to transport over long distances or through dense understory (Table 1).

Additionally, the corrugated plastic targets required rotation every time the scanner moved positions, similar to

the Leica targets, which added extra time to complete each site. A site with nine scans would take approximately 2 h to complete, or about 13 min per scan. We also found that if targets were not placed perfectly, or when wind exposure was high, the likelihood of target movement during or be-

Table 1. Comparison of three different target configurations.

Type of target configurations	Cost per tripod (CAD)	Cost per target (CAD)	Total cost per configuration (CAD)	Collapsed width (m)	Weight per unit (kg)	Weight per person** (kg)	Works in all terrains?	Auto-align?
Leica (L)	\$550.00	\$440.00	\$990.00	0.13	3.83	15.32	Yes	No
Land survey (LS)	\$120.00*	\$2.00	\$122.00*	0.13	4.55	18.2	Yes	No
Monopod (MP)	\$22.99	\$4.83	\$27.82	0.03	0.61	2.44	No	Yes

*Includes wood, screw clamp, and steel telescopic pole.

**Assuming a team of two and eight target stands.

tween scans increased, which posed a challenge during scan alignment.

Monopods

The MP design consisted of a telescopic aluminum pole atop a wooden stake, and a polystyrene spherical target (Fig. 3c). To ensure their stability, the poles were driven into the ground until their stakes were buried. This target stand was the most cost-effective option. The aluminum telescopic pole was \$22.99 CAD purchased from Canadian Tire (www.canadiantire.ca) and a polystyrene sphere target was \$4.83 CAD purchased from Amazon (www.amazon.ca). This configuration was also very light, with a total weight of 0.61 kg. Their lightweight design made them relatively easy to transport through dense understory vegetation and deploy quickly. They were also the most compact option, with a width of 0.03 m. This configuration was also easy to acquire and replace if damaged. However, ease of placement varied significantly by terrain. For instance, in rocky or densely vegetated areas, stability was more challenging to achieve and would require additional support (Table 1).

The polystyrene sphere targets offered additional advantages. They were more easily detected when partially occluded by vegetation or other obstacles compared to traditional B&W targets. The Leica Cyclone REGISTER 360 PLUS (BLK edition) software was able to detect the target even when only a small part of the sphere was captured in the point cloud. We also found that the auto-alignment feature on the scanner was more accurate when used with spheres: if two scans could see at least three of the same targets, then they could be aligned automatically by the Leica software. Finally, the sphere did not need to be moved or rotated during the entire multiscan procedure, which saved time and effort. An average scan would take 10 min, for a total of 1.5 h per site.

Target positional precision

There was significant variation between the target configurations' target position RMS (ANOVA; $F_{(2, 442)} = 34.2, p < 0.001$; Fig. 6). The post hoc Tukey's test with 0.05 alpha showed that the mean RMS for the MP configuration ($M = 0.011$, $SD = 0.017$) was significantly lower than that of the L configuration ($M = 0.020$, $SD = 0.019$), which was itself significantly lower than that of the LS configuration ($M = 0.035$, $SD = 0.041$). The MP design with the spherical targets was therefore the most precise target configuration, followed by the L design.

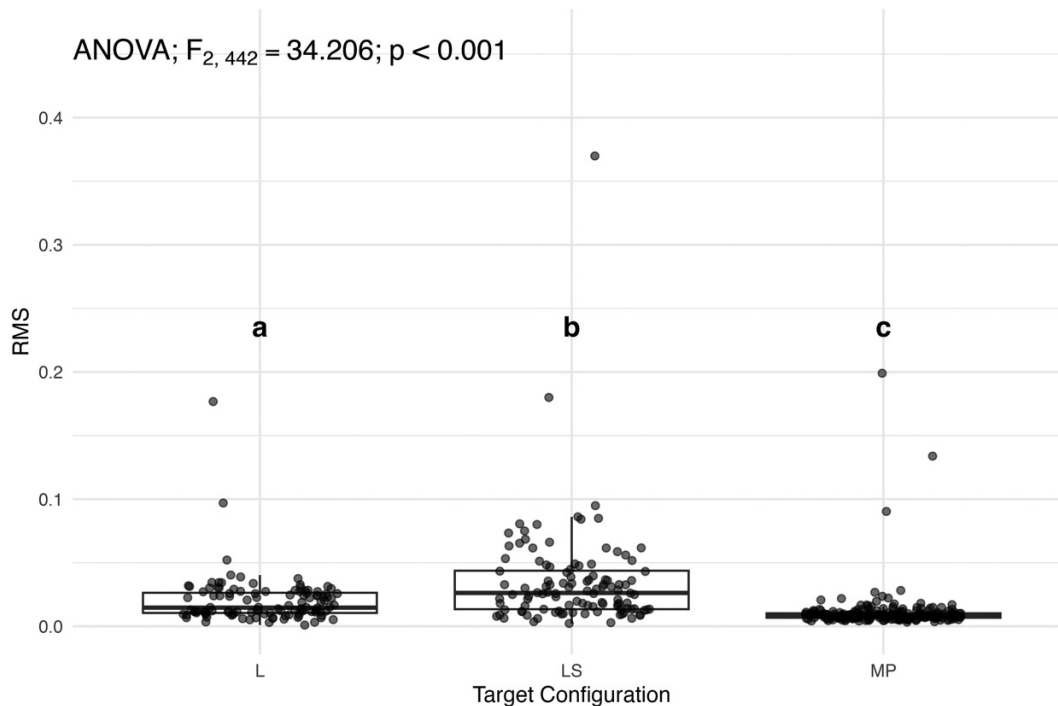
Discussion

The goal of this research note was to investigate alternative target configuration options to be used with TLS in forestry-related field work. We found that polystyrene spheres on monopod stands are an excellent solution for target and support selection in multiscan TLS forestry protocols. We compared three different target configurations in the field for TLS scanning in terms of their cost, weight, ease of use, and precision both in the field and in postprocessing.

The two tripod systems had numerous advantages. The L configuration's high-quality construction made it easy to deploy and was useful across various terrains. The LS configuration was useful on various terrain and was much cheaper than the L configuration. Although the L tripod was nice to work with, both tripod systems were heavy and bulky. This meant that the field team required more personnel to transport materials to the sites. It also significantly increased the physical strain on the field team, which meant that data collection took more days to complete, all else being equal. Despite this, both tripods were relatively durable in the field. For the L configuration, both the targets and tripods held up well over two extensive field seasons; however, they were more difficult for field teams to repair or replace. For example, locating or ordering specific parts in more remote field locations can be challenging with high-quality specialty equipment. The LS tripods remained intact; however, the targets often needed to be replaced. These field repairs and replacements were easier to accomplish as parts were readily available.

The MP configuration was much lighter and more compact than both the L and LS tripod options, improving the experience of personnel, especially during remote or long sampling studies. The polystyrene spheres save considerable time during the scanning procedure, since they do not need to be rotated when the scanner is moved through the site. We found that using the MP configuration saved on average 30 min per plot, allowing more sites to be completed in a day. While all target configurations showed relatively high precision, the spherical targets were significantly more precise compared to the L and LS designs (Fig. 6), which saved additional time during postprocessing as the alignment could be done automatically in most situations, once the targets are registered. This also introduced less human error by avoiding the need for visual scan alignment. Higher target positional precision in the MP configuration indicates better concordance among merged point clouds, ultimately suggesting better data quality. Sphere targets are also less affected by occlusion, as even

Fig. 6. Boxplots showing the root mean square (RMS), a measure of precision, of target positions after alignment for each target configuration. Each point represents a target (L: $n = 112$; LS: $n = 112$; MP: $n = 221$) captured by at least two different scans in sites with sparse understory. All target type's RMS distributions are significantly different from each other (ANOVA; $F_{2, 442} = 34.206$; $p < 0.001$) following a Tukey's test ($\alpha = 0.05$). Higher target positional precision in the MP configuration compared to the L and LS configurations indicates better concordance among merged point clouds.



a small section of visible target will be recognized by the software. Additionally, both spheres and monopods are relatively affordable and accessible, allowing easy acquisition and replacement if needed. Unexpectedly, we found the monopod and polystyrene spheres to be very durable, showing no significant signs of wear after an entire fieldwork season. Additionally, replacing damaged or worn-out components was relatively easy as most parts are readily available. The use of monopods and polystyrene sphere targets greatly reduced cost and personnel barriers to using TLS in forestry.

The presented MP design requires a stake to be driven into the ground. As a result, the monopods require a top-soil layer that is deep enough for the stake to stay in place and prevent the target from moving during the scanning process. We found that when the soil was too shallow or too compact, the monopods were more likely to move during or between scans. This could be mitigated by using additional supports at the base of the monopod, such as extra stakes to prevent movement. The tripod configurations did not encounter this issue. The L configuration was the most secure and could be set up on all terrains. The LS design was also stable. However, the corrugated plastic target was more likely to move in windier conditions. In cases where the soil is too shallow or the sites have exposed bedrock, which is common in the Boreal shield, tripods might be more appropriate than monopods. However, we recommend that spherical targets still be used with the tripod configurations to benefit from not rotating targets between scans and the automatic alignment. In addition, the LS design could be adapted to use the lighter aluminum telescopic poles as used in the MP design.

This would provide added stability with a lower weight and cost and ensure that a tripod option is available if needed.

Forestry field work can be very demanding, requiring expensive resources, durable equipment, personnel, and expertise. We have found that the MP design can greatly reduce strain on field teams and supervisors, while marginally improving data quality. The MP configuration reduces the cost and weight associated with TLS scanning, making this forest stand analysis method more accessible for researchers and land managers. This method also reduces the number of people required in the field for carrying equipment and reduces the time to complete the multiscan protocol by removing the need to turn targets. However, in study sites with shallow organic layers or exposed bedrock, field teams should consider the use of tripods with the polystyrene spherical target.

Terrestrial lidar is becoming a more affordable and accessible technology. While typically deployed for understanding built environments, more forestry research is adopting TLS due to its large amount of accurate and reproducible data. However, in forests, the TLS system can encounter challenges such as scan occlusion, heavy equipment, and high cost. The multiscan approach that is being adopted by many forestry teams (Boucher et al. 2022) requires the use of targets and tripods to merge multiple scans. Since forestry field work is often conducted in remote and hard-to-access locations, streamlining equipment weight and bulk is necessary. Further reducing barriers to TLS uptake can be achieved by reducing equipment costs and weight, which ensures that the protocol can be implemented by more teams. Forestry researchers should consider equipment needs during their

field planning to best optimize data collection and field team morale.

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Data availability

This manuscript does not report data.

Author information

Author ORCIDs

Kennedy Korkola <https://orcid.org/0009-0004-6079-783X>

Author notes

Léo Jourdan and Kennedy Korkola contributed equally to the paper.

Author contributions

Conceptualization: LJ, KK, PP

Funding acquisition: PMAJ

Investigation: LJ, KK

Project administration: PMAJ

Resources: LJ, KK, PP, PMAJ

Supervision: PMAJ

Writing – original draft: LJ, KK

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Competing interests

The authors declare there are no competing interests.

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